AERIS Expert Review Packet

Anomaly

Source / metric	openaq / ozone
Observed / expected	0.088 / 0.0166316
Time	2026-07-22 19:00 UTC
Location	29.5831, -95.0156
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	6 / 667	29.01 to 100; mean 71.2211	44.81 at 2026-07-22 18:55Z; dt 5 min
asos	temperature	degC	6 / 667	23.2222 to 39; mean 29.9494	37 at 2026-07-22 18:55Z; dt 5 min
asos	wind_direction	degree	6 / 759	0 to 360; mean 175.837	310 at 2026-07-22 18:55Z; dt 5 min
asos	wind_speed	m/s	6 / 783	0 to 8.74555; mean 2.9454	3.08666 at 2026-07-22 18:55Z; dt 5 min
noaa_gfs	gh_500	m	6 / 72	5915.17 to 5959.62; mean 5933.16	5935.62 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	pbl_height	m	6 / 72	17.7446 to 2017.14; mean 587.765	1612.34 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	precipitable_water	mm	6 / 72	29.7729 to 67.4723; mean 49.8948	55.941 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	surface_pressure	hPa	6 / 72	1004.59 to 1014.29; mean 1009.09	1009.5 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	t_850	degC	6 / 72	17.875 to 24.3609; mean 21.2647	21.7781 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	u_10m	m/s	6 / 72	-4.5306 to 4.0994; mean 0.7762	0.121985 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	v_10m	m/s	6 / 72	-6.0398 to 5.78799; mean -0.288	-2.02513 at 2026-07-22 18:00Z; dt 60 min
openaq	ozone	ppm	15 / 1011	-0.004 to 0.092; mean 0.0274	0.088 at 2026-07-22 19:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 178	3 to 98; mean 37.0449	73 at 2026-07-22 19:00Z; dt 0 min
openaq	pm25	ug/m3	64 / 2493	0 to 34.1; mean 9.3529	10.2 at 2026-07-22 19:00Z; dt 0 min
openweather	cloud_cover	percent	3 / 216	0 to 100; mean 47.1944	17 at 2026-07-22 19:01Z; dt 1.9 min
openweather	humidity	percent	3 / 216	43 to 96; mean 73.7407	50 at 2026-07-22 19:01Z; dt 1.9 min
openweather	precipitation	mm/h	3 / 216	0 to 11.53; mean 0.2961	0 at 2026-07-22 19:01Z; dt 1.9 min
openweather	pressure	hPa	3 / 216	1007 to 1015; mean 1010.44	1009 at 2026-07-22 19:01Z; dt 1.9 min
openweather	temperature	degC	3 / 216	24.14 to 40.07; mean 30.6104	38.57 at 2026-07-22 19:01Z; dt 1.9 min
openweather	wind_direction	degree	3 / 216	0 to 360; mean 189.287	26 at 2026-07-22 19:01Z; dt 1.9 min
openweather	wind_speed	m/s	3 / 216	0.4 to 7.61; mean 2.6884	3.6 at 2026-07-22 19:01Z; dt 1.9 min
purpleair	pm25	ug/m3	59 / 4244	0 to 149.3; mean 11.4354	6.8 at 2026-07-22 19:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; dt 14.2 min
sentinel5p	s5p_ch4_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; dt 14.2 min
sentinel5p	s5p_co_column	mol/m^2	6 / 6	0.0233633 to 0.0312825; mean 0.0271	0.0312825 at 2026-07-22 18:45Z; dt 14.2 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; dt 14.2 min
sentinel5p	s5p_hcho_column	mol/m^2	5 / 5	0.000163343 to 0.000396712; mean 0.0002	0.000396712 at 2026-07-22 19:14Z; dt 14.6 min
sentinel5p	s5p_hcho_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-22 19:14Z; dt 14.6 min
sentinel5p	s5p_no2_column	mol/m^2	2 / 2	5.0358e-05 to 5.87309e-05; mean 0.0001	5.0358e-05 at 2026-07-22 19:14Z; dt 14.6 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-07-22 19:14Z; dt 14.6 min
sentinel5p	s5p_o3_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; dt 14.2 min
sentinel5p	s5p_so2_column	mol/m^2	5 / 5	-2.41161e-06 to 9.61895e-05; mean 0	-2.41161e-06 at 2026-07-22 19:14Z; dt 14.6 min
sentinel5p	s5p_so2_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-22 19:14Z; dt 14.6 min
tceq	co	ppm	3 / 195	0 to 0.5; mean 0.1436	0.1 at 2026-07-22 19:00Z; dt 0 min
tceq	no2	ppb	11 / 671	-2.6 to 44.3; mean 6.3422	5.4 at 2026-07-22 19:00Z; dt 0 min
tceq	so2	ppb	3 / 191	-1 to 1.6; mean -0.2068	1.2 at 2026-07-22 19:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

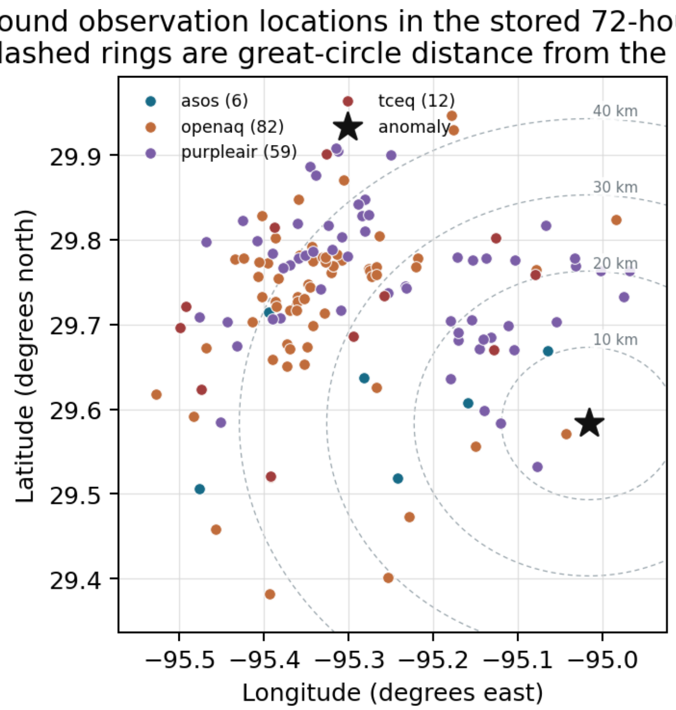
Source	Metric	Values
asos	humidity	07-21 06Z 93.19, 12Z 73.74, 18Z 39.69, 07-22 00Z 60.95, 06Z 79.14, 12Z 59.63, 18Z 42.15, 07-23 00Z 70.41, 06Z 75.71, 12Z 74.58, 18Z 86.64, 07-24 00Z 92.69, 06Z 87.93
asos	temperature	07-21 06Z 25.03, 12Z 29.02, 18Z 35.4, 07-22 00Z 30.81, 06Z 27.01, 12Z 31.64, 18Z 37.78, 07-23 00Z 31.34, 06Z 29.2, 12Z 29.13, 18Z 27.92, 07-24 00Z 25.98, 06Z 25.79
asos	wind_direction	07-21 06Z 183.4, 12Z 261.5, 18Z 218.6, 07-22 00Z 201.6, 06Z 237, 12Z 289.8, 18Z 212.8, 07-23 00Z 118.9, 06Z 45, 12Z 58.03, 18Z 159.8, 07-24 00Z 151.4, 06Z 108.2
asos	wind_speed	07-21 06Z 2.077, 12Z 3.508, 18Z 2.36, 07-22 00Z 2.476, 06Z 2.982, 12Z 3.928, 18Z 2.666, 07-23 00Z 1.972, 06Z 3, 12Z 4.49, 18Z 2.557, 07-24 00Z 3.219, 06Z 2.432
noaa_gfs	gh_500	07-21 12Z 5916, 18Z 5931, 07-22 00Z 5929, 06Z 5934, 12Z 5926, 18Z 5938, 07-23 00Z 5927, 06Z 5941, 12Z 5925, 18Z 5933, 07-24 00Z 5940, 06Z 5957
noaa_gfs	pbl_height	07-21 12Z 184.9, 18Z 1188, 07-22 00Z 545.7, 06Z 109.9, 12Z 219.3, 18Z 1663, 07-23 00Z 314.4, 06Z 306.1, 12Z 329.8, 18Z 1188, 07-24 00Z 528.8, 06Z 475.5
noaa_gfs	precipitable_water	07-21 12Z 31.33, 18Z 35.78, 07-22 00Z 40.72, 06Z 45.72, 12Z 52.55, 18Z 55.8, 07-23 00Z 61.66, 06Z 46.98, 12Z 51.37, 18Z 64.06, 07-24 00Z 63.55, 06Z 49.23
noaa_gfs	surface_pressure	07-21 12Z 1011, 18Z 1011, 07-22 00Z 1008, 06Z 1009, 12Z 1009, 18Z 1009, 07-23 00Z 1006, 06Z 1008, 12Z 1008, 18Z 1008, 07-24 00Z 1009, 06Z 1013
noaa_gfs	t_850	07-21 12Z 22.46, 18Z 22.28, 07-22 00Z 21.85, 06Z 22.4, 12Z 21.68, 18Z 21.72, 07-23 00Z 23.75, 06Z 21.61, 12Z 20.18, 18Z 19.34, 07-24 00Z 19.88, 06Z 18.04
noaa_gfs	u_10m	07-21 12Z 2.739, 18Z 2.308, 07-22 00Z 0.8438, 06Z 2.517, 12Z 3.085, 18Z 0.662, 07-23 00Z -0.1673, 06Z -1.059, 12Z -0.3388, 18Z -1.688, 07-24 00Z 1.223, 06Z -0.8123

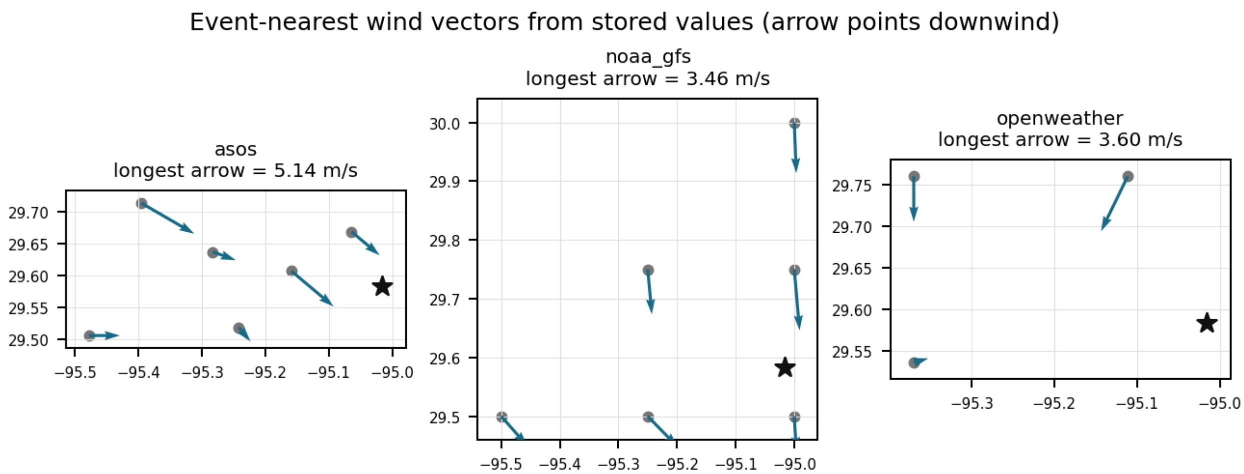
Source	Metric	Values
noaa_gfs	v_10m	07-21 12Z 0.2999, 18Z -0.7365, 07-22 00Z 0.5354, 06Z 1.391, 12Z -0.4914, 18Z -2.383, 07-23 00Z 1.481, 06Z -3.055, 12Z -3.632, 18Z -5.125, 07-24 00Z 4.78, 06Z 3.481
openaq	ozone	07-21 06Z 0.006118, 12Z 0.01152, 18Z 0.04518, 07-22 00Z 0.02902, 06Z 0.00619, 12Z 0.01574, 18Z 0.06606, 07-23 00Z 0.03856, 06Z 0.02697, 12Z 0.02675, 18Z 0.02681, 07-24 00Z 0.02054, 06Z 0.01633
openaq	pm10	07-21 06Z 39.3, 12Z 41.33, 18Z 49.09, 07-22 00Z 46.83, 06Z 42.2, 12Z 45.33, 18Z 51.67, 07-23 00Z 55, 06Z 30.56, 12Z 25.22, 18Z 16.39, 07-24 00Z 19.61, 06Z 36.67
openaq	pm25	07-21 06Z 7.323, 12Z 6.929, 18Z 5.735, 07-22 00Z 5.895, 06Z 6.012, 12Z 6.28, 18Z 9.156, 07-23 00Z 10.76, 06Z 17.85, 12Z 17.49, 18Z 8.554, 07-24 00Z 8.028, 06Z 13.58
openweather	cloud_cover	07-21 06Z 0, 12Z 0.7059, 18Z 1.474, 07-22 00Z 4.389, 06Z 25.94, 12Z 13.06, 18Z 63.39, 07-23 00Z 80.88, 06Z 74.79, 12Z 99.22, 18Z 99, 07-24 00Z 95.06, 06Z 78.5
openweather	humidity	07-21 06Z 90.94, 12Z 80.12, 18Z 48.68, 07-22 00Z 62.89, 06Z 80.11, 12Z 70.67, 18Z 49.17, 07-23 00Z 74.12, 06Z 77.89, 12Z 78.33, 18Z 80, 07-24 00Z 93.28, 06Z 93
openweather	precipitation	07-21 06Z 0, 12Z 0, 18Z 0, 07-22 00Z 0, 06Z 0, 12Z 0, 18Z 0.01, 07-23 00Z 2.265, 06Z 0, 12Z 0, 18Z 0.6411, 07-24 00Z 0.6922, 06Z 0.635
openweather	pressure	07-21 06Z 1012, 12Z 1013, 18Z 1011, 07-22 00Z 1010, 06Z 1010, 12Z 1011, 18Z 1008, 07-23 00Z 1009, 06Z 1009, 12Z 1010, 18Z 1009, 07-24 00Z 1012, 06Z 1014
openweather	temperature	07-21 06Z 25.58, 12Z 28.88, 18Z 36.56, 07-22 00Z 31.93, 06Z 27.36, 12Z 31.33, 18Z 39.05, 07-23 00Z 31.87, 06Z 29.81, 12Z 29.42, 18Z 29.4, 07-24 00Z 25.86, 06Z 25.3
openweather	wind_direction	07-21 06Z 179.7, 12Z 268.3, 18Z 255.6, 07-22 00Z 224.1, 06Z 258.9, 12Z 283.7, 18Z 172.4, 07-23 00Z 221, 06Z 89.32, 12Z 63.67, 18Z 99.06, 07-24 00Z 165.2, 06Z 167
openweather	wind_speed	07-21 06Z 2.419, 12Z 2.092, 18Z 1.464, 07-22 00Z 2.726, 06Z 2.276, 12Z 2.136, 18Z 2.466, 07-23 00Z 2.711, 06Z 2.738, 12Z 3.544, 18Z 4.686, 07-24 00Z 2.944, 06Z 3.245
purpleair	pm25	07-21 06Z 9.591, 12Z 8.39, 18Z 7.889, 07-22 00Z 9.967, 06Z 6.817, 12Z 7.95, 18Z 11.08, 07-23 00Z 14.7, 06Z 22.34, 12Z 17.32, 18Z 9.85, 07-24 00Z 10.33, 06Z 15.13
sentinel5p	s5p_aer_ai_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_ch4_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_co_column	07-21 18Z 0.02656, 07-22 18Z 0.03123, 07-23 18Z 0.02345
sentinel5p	s5p_co_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_hcho_column	07-21 18Z 0.0002243, 07-22 18Z 0.0003967, 07-23 18Z 0.0001668
sentinel5p	s5p_hcho_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_no2_column	07-21 18Z 5.873e-05, 07-22 18Z 5.036e-05
sentinel5p	s5p_no2_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_o3_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_so2_column	07-21 18Z 3.63e-05, 07-22 18Z -2.412e-06, 07-23 18Z 5.971e-05
sentinel5p	s5p_so2_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
tceq	co	07-21 06Z 0.1267, 12Z 0.1556, 18Z 0.1056, 07-22 00Z 0.21, 06Z 0.1056, 12Z 0.1556, 18Z 0.1222, 07-23 00Z 0.18, 06Z 0.1333, 12Z 0.1333, 18Z 0.1611, 07-24 00Z 0.22, 06Z 0.1167
tceq	no2	07-21 06Z 5.508, 12Z 5.514, 18Z 4.918, 07-22 00Z 10.8, 06Z 6.498, 12Z 7.011, 18Z 6.311, 07-23 00Z 7.194, 06Z 5.151, 12Z 6.013, 18Z 6.791, 07-24 00Z 7.521, 06Z 4.65
tceq	so2	07-21 06Z -0.28, 12Z -0.275, 18Z -0.1611, 07-22 00Z -0.1222, 06Z -0.2889, 12Z -0.07222, 18Z 0.07778, 07-23 00Z -0.1667, 06Z -0.3, 12Z -0.2333, 18Z -0.3278, 07-24 00Z -0.27, 06Z -0.35

Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	T41 (10.66 km) 72.29, EFD (14.095 km) 70.67, LVJ (23.006 km) 75.3, HOU (26.492 km) 71.27, MCJ (39.448 km) 67.2, AXH (45.438 km) 71.91
asos	temperature	T41 (10.66 km) 29.96, EFD (14.095 km) 30.31, LVJ (23.006 km) 29.3, HOU (26.492 km) 30.06, MCJ (39.448 km) 30.59, AXH (45.438 km) 29.46
asos	wind_direction	T41 (10.66 km) 190.3, EFD (14.095 km) 180.9, LVJ (23.006 km) 165.3, HOU (26.492 km) 183.1, MCJ (39.448 km) 197.6, AXH (45.438 km) 144.1
asos	wind_speed	T41 (10.66 km) 3.183, EFD (14.095 km) 3.451, LVJ (23.006 km) 1.993, HOU (26.492 km) 3.58, MCJ (39.448 km) 4.13, AXH (45.438 km) 1.612
noaa_gfs	gh_500	gfs:29.50,-95.00 (9.363 km) 5931, gfs:29.75,-95.00 (18.62 km) 5934, gfs:29.50,-95.25 (24.486 km) 5931, gfs:29.75,-95.25 (29.28 km) 5934, gfs:30.00,-95.00 (46.382 km) 5936, gfs:29.50,-95.50 (47.763 km) 5932
noaa_gfs	pbl_height	gfs:29.50,-95.00 (9.363 km) 538.8, gfs:29.75,-95.00 (18.62 km) 591.1, gfs:29.50,-95.25 (24.486 km) 556.4, gfs:29.75,-95.25 (29.28 km) 863.1, gfs:30.00,-95.00 (46.382 km) 434.7, gfs:29.50,-95.50 (47.763 km) 542.5
noaa_gfs	precipitable_water	gfs:29.50,-95.00 (9.363 km) 50.8, gfs:29.75,-95.00 (18.62 km) 50.18, gfs:29.50,-95.25 (24.486 km) 50.31, gfs:29.75,-95.25 (29.28 km) 49.3, gfs:30.00,-95.00 (46.382 km) 49.81, gfs:29.50,-95.50 (47.763 km) 48.96
noaa_gfs	surface_pressure	gfs:29.50,-95.00 (9.363 km) 1010, gfs:29.75,-95.00 (18.62 km) 1010, gfs:29.50,-95.25 (24.486 km) 1009, gfs:29.75,-95.25 (29.28 km) 1009, gfs:30.00,-95.00 (46.382 km) 1008, gfs:29.50,-95.50 (47.763 km) 1008
noaa_gfs	t_850	gfs:29.50,-95.00 (9.363 km) 21.12, gfs:29.75,-95.00 (18.62 km) 21.3, gfs:29.50,-95.25 (24.486 km) 21.15, gfs:29.75,-95.25 (29.28 km) 21.33, gfs:30.00,-95.00 (46.382 km) 21.44, gfs:29.50,-95.50 (47.763 km) 21.25
noaa_gfs	u_10m	gfs:29.50,-95.00 (9.363 km) 1.151, gfs:29.75,-95.00 (18.62 km) 0.3923, gfs:29.50,-95.25 (24.486 km) 0.7273, gfs:29.75,-95.25 (29.28 km) 0.4448, gfs:30.00,-95.00 (46.382 km) 0.4023, gfs:29.50,-95.50 (47.763 km) 1.54
noaa_gfs	v_10m	gfs:29.50,-95.00 (9.363 km) 0.4249, gfs:29.75,-95.00 (18.62 km) -0.1517, gfs:29.50,-95.25 (24.486 km) -0.1709, gfs:29.75,-95.25 (29.28 km) -0.5792, gfs:30.00,-95.00 (46.382 km) -0.9509, gfs:29.50,-95.50 (47.763 km) -0.3001
openaq	ozone	5077791 (0.0 km) 0.02484, 332 (14.578 km) 0.03371, 2800 (21.045 km) 0.02357, 320 (24.779 km) 0.02815, 3214 (26.61 km) 0.02791, 339 (26.887 km) 0.02982, 2039 (28.555 km) 0.02512, 3378 (28.767 km) 0.02825, +7 more
openaq	pm10	271 (14.578 km) 44.44, 13380082 (28.767 km) 34.43, 15852419 (35.779 km) 30.83
openaq	pm25	14404954 (2.966 km) 9.159, 13955815 (13.303 km) 7.753, 268 (14.578 km) 15.92, 11638043 (23.947 km) 8.072, 14157975 (24.782 km) 9.895, 3379 (28.767 km) 10.63, 15539682 (28.768 km) 10.16, 8967123 (28.822 km) 11.02, +56 more
openweather	cloud_cover	grid:east (21.767 km) 44.26, grid:south (34.66 km) 50.92, grid:center (39.494 km) 46.4
openweather	humidity	grid:east (21.767 km) 77.22, grid:south (34.66 km) 73.29, grid:center (39.494 km) 70.71
openweather	precipitation	grid:east (21.767 km) 0.2751, grid:south (34.66 km) 0.404, grid:center (39.494 km) 0.2092
openweather	pressure	grid:east (21.767 km) 1010, grid:south (34.66 km) 1010, grid:center (39.494 km) 1010
openweather	temperature	grid:east (21.767 km) 30.13, grid:south (34.66 km) 30.36, grid:center (39.494 km) 31.35
openweather	wind_direction	grid:east (21.767 km) 183.1, grid:south (34.66 km) 188.1, grid:center (39.494 km) 196.7
openweather	wind_speed	grid:east (21.767 km) 3.235, grid:south (34.66 km) 2.457, grid:center (39.494 km) 2.373
purpleair	pm25	2386 (8.161 km) 0.5181, 3298 (10.104 km) 12.59, 247283 (12.088 km) 12.7, 268165 (12.918 km) 13.56, 222815 (13.885 km) 11.74, 222593 (15.857 km) 12.19, 128007 (15.88 km) 6.247, 161689 (15.935 km) 12.4, +51 more
tceq	co	482011039 (14.558 km) 0.09538, 482011035 (28.764 km) 0.1672, 482011052 (44.204 km) 0.1682
tceq	no2	482011050 (0.008 km) 4.117, 482011039 (14.558 km) 7.495, 482011015 (20.5 km) 8.597, 482010026 (26.626 km) 7.075, 482011035 (28.764 km) 7.012, 482010416 (29.317 km) 6.567, 480391004 (37.118 km) 4.888, 482011052 (44.204 km) 3.948, +3 more
tceq	so2	482011039 (14.558 km) 0.3185, 482011035 (28.764 km) -0.4794, 482010051 (44.584 km) -0.4762


Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
asos (6)
openaq (82)
purpleair (59)
tceq (12)
anomaly
40 km
30 km
20 km
10 km
Latitude (degrees north)
Longitude (degrees east)
29.9
29.8
29.7
29.6
29.5
29.4
−95.5
−95.4
−95.3
−95.2
−95.1
−95.0


Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 5.14 m/s
noaa_gfs
longest arrow = 3.46 m/s
openweather
longest arrow = 3.60 m/s

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

An ASOS weather station 10.66 km from the anomaly recorded a high surface temperature of 37.0 degC and relative humidity of 44.81 percent at 18:55:00 UTC on July 22, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

Elevated concentrations of volatile organic compound precursors, evidenced by elevated Sentinel-5P formaldehyde column density measurements, provided the necessary chemical reactivity to drive rapid tropospheric ozone formation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

Hot, dry, sunny afternoon conditions were strongly favorable for rapid surface-ozone production near eastern Houston at the anomaly time, because air temperature was about 37 to 39 degrees C, relative humidity was about 45 to 50 percent, cloud cover near the anomaly was only 17 percent, and no precipitation was occurring.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

Light surface winds and an expanding afternoon planetary boundary layer exceeding 1600 meters promoted local precursor accumulation and thermal mixing that enhanced afternoon ozone production.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

Transport or recirculation of urban-industrial precursor emissions into eastern Houston was a likely contributor to the ozone spike, because winds were variable but included northwesterly flow at the nearest surface station near the event while satellite columns showed elevated carbon monoxide and strongly elevated formaldehyde with no corresponding sulfur dioxide enhancement.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

Chemical observations near eastern Houston on 2026-07-22 are most consistent with VOC-rich photochemical ozone production and transport because ozone reached 0.088 ppm while satellite formaldehyde and carbon monoxide were elevated, local nitrogen dioxide was only moderate, sulfur dioxide was not elevated, and particulate matter did not show a major smoke signature.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

Satellite measurements detected a peak in atmospheric formaldehyde column density of 0.000397 mol/m² on July 22, 2026, pointing to strong organic precursor availability during the event.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

The PM2.5 levels are elevated at 6.8 ug/m3, which is higher than the mean of 11.4354 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

The air quality monitoring data from multiple sources, including PurpleAir and TCEQ, confirmed the anomaly and provided valuable insights into its characteristics.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

Boundary-layer structure near eastern Houston at 18:00 UTC on 2026-07-22 favored efficient entrainment and mixing of ozone and its precursors because the planetary boundary layer height was about 1612 m under relatively high 500 hPa heights.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

High surface temperatures exceeding 37 degC combined with clear conditions on July 22, 2026, significantly accelerated photochemical reaction rates to yield elevated ground-level ozone concentrations in the Houston area.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

A deep planetary boundary layer height reaching approximately 1663 meters near peak sunlight facilitated the vertical mixing of photochemical precursors over the Houston region.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

The 6-hour mean openaq ozone concentration for 2026-07-22 18Z was 0.06606 ppm, which was substantially higher than the preceding 2026-07-22 12Z mean of 0.01574 ppm within the 50 km context window.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

Sentinel-5P satellite observations detected an elevated formaldehyde (HCHO) total column density of 0.000397 mol/m^2 over the region on July 22, 2026, at 19:14:34 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

Deep daytime mixing likely supported the ozone spike near eastern Houston by distributing ozone and precursors through a well-developed boundary layer, because modeled planetary boundary-layer height near the site was about 1612 m at 18Z on 2026-07-22 and 500 hPa heights were relatively high near 5936 m.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

Numerical weather model data indicated a deep planetary boundary layer height exceeding 1,600 meters on the afternoon of July 22, 2026, supporting intense daytime boundary layer development and photochemical reaction volume.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

Surface meteorology near eastern Houston at 19:00 UTC on 2026-07-22 strongly favored rapid ozone formation because air temperature was about 37 to 39 C, relative humidity was about 45 to 50 percent, cloud cover was low near the anomaly time, and no precipitation was occurring.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

The air quality anomaly was characterized by elevated levels of $PM_{2.5}$ and NO_2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

Strong photochemical ozone production in eastern Houston was likely enhanced on 2026-07-22 by very hot afternoon temperatures near 37 to 39 degrees Celsius, relatively low humidity near 45 to 50 percent, little precipitation, and low cloud cover near the anomaly time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

A deep and well-mixed afternoon boundary layer over Houston on 2026-07-22 likely promoted rapid ozone formation and downward mixing of ozone-rich air, because the planetary boundary layer height was about 1612 meters near 18Z under elevated 500 hPa heights.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

High ambient temperatures exceeding 37 degrees Celsius combined with low relative humidity created ideal atmospheric conditions for rapid photochemical ozone production in Houston on July 22, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to poor air mixing and increased pollutant concentrations.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

Transport or recirculation of VOC-rich urban-industrial emissions is supported more strongly than a fresh sulfur-rich or combustion-dominated plume, because Sentinel-5P formaldehyde and carbon monoxide columns were elevated on 2026-07-22 while NO_2 near the surface was only modest at 5.4 ppb, TCEQ CO was low at 0.1 ppm, and both Sentinel-5P SO_2 and surface SO_2 remained low near the anomaly time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

The anomaly was likely caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

Satellite observations revealed a notable spike in atmospheric formaldehyde column density on July 22, 2026, providing evidence of elevated volatile organic compound concentrations that acted as ozone precursors.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

The openaq ozone anomaly at (29.583, -95.016) had an expected value of 0.016631578947368424 ppm and a z-score of 5.812520004185725, which is labeled severe in the provided data.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO_2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

The anomalous pollutant was ozone measured by openaq at 0.088 ppm at 2026-07-22T19:00:00+00:00 at coordinates (29.583, -95.016).

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

Near-surface atmospheric temperatures reached up to 39.0°C to 40.07°C alongside relative humidity dropping below 45% around 19:00 UTC on July 22, 2026, providing thermal conditions conducive to photochemical ozone production.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

The anomaly occurred on July 22nd at around 19:00 UTC, with the nearest sensor located approximately 37.201 km away.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

An OpenAQ sensor recorded a severe ozone anomaly of 0.088 ppm at coordinates (29.583, -95.016) on July 22, 2026, at 19:00:00 UTC, compared to an expected baseline of approximately 0.0166 ppm (z-score 5.81).

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.